Summary in Graph

# Exam Summary (GO Classes CS Test Series 2026 | Engineering Mathematics | Subject Wise Test 2)

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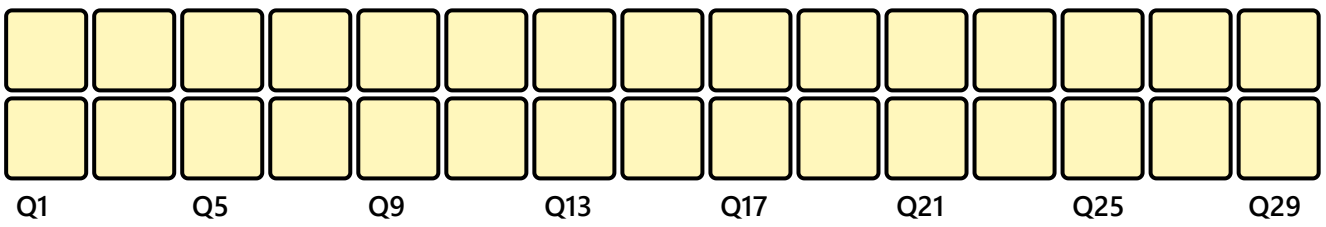
|---|---|---|---|
| Qs. Attempted: | 0 (0 + 0) | Correct Marks: | 0 (0 + 0) |
| Correct Attempts: | 0 (0 + 0) | Penalty Marks: | 0 (0 + 0) |
| Incorrect Attempts: | 0 (0 + 0) | Resultant Marks: | 0 (0 + 0) |

| | |
|---|---|
| Total Questions: | 30 (20 + 10) |
| Total Marks: | 40 (20 + 20) |
| Exam Duration: | 90 Minutes |
| Time Taken: | 0 Minutes |

## Individual Question Statistics

Select Metric: Marks Obtained




* Average Stats based on first-attempt stats of Top 10% (29) out of 290 users

EXAM RESPONSE | EXAM STATS | FEEDBACK

## Technical

Q #1 | Numerical Type | Award: 1 | Penalty: 0 | Calculus

Evaluate the limit:

$$\lim_{x \to 0} \frac{\frac{1}{3}x^3 + 2x - 2\sin x}{4x^3} = \frac{1}{a}$$

Find the value of $\mathbf{a}$.

Your Answer: | Correct Answer: 6 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

The value of $\int_{-1}^{1} |x| \mathrm{d}x$ is :

A. $-2$

B. $-1$

C. $1$

D. $2$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

Which of the following statements is true for the function $\mathrm{f(x)} = \begin{cases} \mathrm{x}^2+3, & \mathrm{x} \neq 0 \\ 1, & \mathrm{x} = 0 \end{cases}$ ?

A. $\mathrm{f(x)}$ is continuous and differentiable $\forall \mathrm{x} \in \mathbb{R}$
B. $f(x)$ is continuous $\forall x \in \mathbb{R}$
C. $\mathrm{f(x)}$ is continuous and differentiable $\forall \mathrm{x} \in \mathbb{R} - \{0\}$
D. $\mathrm{f(x)}$ is discontinuous at infinitely many points

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

Suppose $A = \begin{bmatrix} a & b \\ b & d \end{bmatrix}$, with $b \neq 0$. If one of the eigenvectors of $A$ is $\begin{bmatrix} 2 \\ 3 \end{bmatrix}$, then another eigenvector of $A$ is

A. $\begin{bmatrix} 3 \\ 2 \end{bmatrix}$

B. $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$

C. $\begin{bmatrix} -3 \\ 2 \end{bmatrix}$

D. $\begin{bmatrix} -2 \\ 3 \end{bmatrix}$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Select Type | Award: 1 | Penalty: 0 | Linear Algebra

Suppose $A$ is a $3 \times 5$ matrix such that the three vectors

$$\begin{bmatrix}1\\1\\0\\2\\1\end{bmatrix}, \begin{bmatrix}2\\4\\7\\-1\\2\end{bmatrix}, \begin{bmatrix}0\\-2\\-7\\5\\0\end{bmatrix}$$

span $\text{null}(A)$. Select all of the following statements that are True.

A. The rows of $A$ are linearly dependent.

B. The columns of $A$ are linearly dependent.

C. The system $A\vec{x} = \vec{b}$ is consistent for every $\vec{b}$ in $\mathbb{R}^3$.

D. The rank of $A$ is $3$ .

Your Answer: | Correct Answer: A;B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

Assume that the $4 \times 4$ matrix $A$ is row equivalent to the matrix $B$, where

$$B = \begin{bmatrix}1 & 0 & 2 & 0\\0 & 1 & 1 & 0\\0 & 0 & 0 & 1\\0 & 0 & 0 & 0\end{bmatrix}$$

Which of the following statements is true?

A. $B$ is not the reduced row echelon form of $A$.
B. $\det(A) \neq 0$.
C. The null space of $A$ has dimension 3 .
D. If $A\vec{x} = \vec{b}$ is consistent for some $\vec{b} \neq \overrightarrow{0}$, then it has infinitely many solutions.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

Which of the following statements about all $5 \times 5$ matrices is true?

A. $\det(A + B) = \det A + \det B$
B. $\det(-A) = -\det(A)$.
C. $AB = 0$ implies $A = 0$ or $B = 0$.
D. If $\det A = 0$ then some two rows are proportional.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Select Type | Award: 1 | Penalty: 0 | Linear Algebra

Consider the following matrices defined using real numbers $a, b, c$ :

$$A = \begin{bmatrix}1 & a\\1 & 1\end{bmatrix}, \quad B = \begin{bmatrix}1 & b\\1 & 1\end{bmatrix}, \quad C = \begin{bmatrix}1 & c\\1 & 1\end{bmatrix}$$

Each matrix is of the form $\begin{bmatrix} 1 & x \\ 1 & 1 \end{bmatrix}$, where $x \in \mathbb{R}$.

Which of the following statements is/are TRUE?

A. Matrix $A$ has two distinct real eigenvalues if and only if $a > 0$.
B. Matrix $B$ has a repeated real eigenvalue if and only if $b = 0$.
C. Matrix $C$ has complex (non-real) eigenvalues if and only if $c < 0$.
D. The eigenvalues of all three matrices are real for any real values of $a, b, c$.

Your Answer: | Correct Answer: A;B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Numerical Type | Award: 1 | Penalty: 0 | Linear Algebra

A $10 \times 13$ matrix $A$ is given and the homogeneous system $A\vec{x} = \vec{0}$ is transformed to reduced row echelon form. There are $7$ lead variables. How many free variables?

Your Answer: | Correct Answer: 6 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

The values of $a$ for which the matrix $\begin{bmatrix} 6 & 0 & a \\ 0 & 2 & 0 \\ a & 0 & 6 \end{bmatrix}$ is not invertible, are

A. $a = \pm 3$
B. $a = \pm 2$
C. $a = \pm 6$
D. $a = \pm 9$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

Consider the following two statements:
$\mathrm{S1}$: If a matrix has determinant zero, then two of its columns are scalar multiples of each other, or one of the columns is the zero vector.
$\mathrm{S2}$: If $\lambda$ is an eigenvalue of $AA^\top$, then $\lambda$ is necessarily an eigenvalue of $A^\top A$.
Which of the following is correct?

A. $\mathrm{S1}$ is true, $\mathrm{S2}$ is false

B. $\mathrm{S1}$ is false, $\mathrm{S2}$ is true

C. Both $\mathrm{S1}$ and $\mathrm{S2}$ are true

D. Both $\mathrm{S1}$ and $\mathrm{S2}$ are false

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

A system $A\mathbf{x} = \mathbf{b}$ with $m \times 2024$ matrix $A$ and $\mathbf{b} = (1, 0, \ldots, 0)^T$ has a unique solution. Which of the these statements does not follow from these conditions?

A. $\text{rank}(A) = 2024$
B. $m \geq 2024$
C. Columns of $A$ are linearly independent
D. $A$ is invertible

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

Let $B_1, B_2$ be mutually exclusive and exhaustive events, with $P(B_1) = P(B_2) = \frac{1}{2}$. Then for any event $A$ with $P(A \mid B_1) = \frac{1}{4}$ and $P(A \mid B_2) = \frac{3}{4}$, we have

A. $P(A) = \frac{1}{2}$.

B. $P(A) = \frac{1}{4}$.

C. $P(A) = \frac{3}{4}$.

D. $P(A) = \frac{1}{5}$.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

Let $X$ be the number of heads obtained in $40$ independent tosses of a fair coin. Then $X$ is a Binomial random variable with

A. $n = 40, p = 0$.

B. $n = 0.5, p = 40$.

C. $n = 40, p = 0.5$.

D. $n = 100, p = 0$.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

If $B_1$ and $B_2$ are mutually exclusive and exhaustive events, then which of the following always holds?

A. $P(B_1 \cup B_2) = 0$.

B. $P(B_1 \cap B_2) = 0$.

C. $P(B_1) = P(B_2)$.

D. $P(B_1) \neq P(B_2)$.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #16 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

Let $X$ be a discrete random variable with probability mass function:

$$f(x) = \binom{3}{x}\left(\frac{1}{4}\right)^{x}\left(\frac{3}{4}\right)^{3-x}, \quad x = 0, 1, 2, 3$$

What is the expected value (mean) of $X$ ?

A. $\frac{3}{4}$

B. $\frac{9}{16}$

C. $1$

D. $\frac{3}{2}$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #17 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Mathematical Logic

Let $E$ and $F$ be two events such that $0 < P(E) < 1$ and $P(E \mid F) + P\left(E \mid F^c\right) = 1$. Then

A. $E$ and $F$ are mutually exclusive

B. $P\left(E^c \mid F\right) + P\left(E^c \mid F^c\right) = 1$

C. $E$ and $F$ are independent

D. $P(E \mid F) + P\left(E^c \mid F^c\right) = 1$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #18 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

Let $X_1 \sim U(0, 4)$, i.e., a uniform distribution on the interval $[0, 4]$.
Then, the value of

$$P\left(X_1 > 2\right) = ?$$

A. $\frac{1}{4}$

B. $\frac{1}{2}$

C. $\frac{3}{4}$

D. $e^{-1}$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #19 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

Let

$$X \sim \text{Binomial}(n, p)$$

and define the random variable

$$Z = n - X$$

Then the distribution of $Z$ is:

A. Binomial with parameters $(n, p)$
B. Binomial with parameters $(n, 1-p)$
C. Binomial with parameters $(n-1, p)$
D. Binomial with parameters $(n-1, 1-p)$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #20 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

If $A$ and $B$ are two events and the probability $P(B) \neq 1$, then evaluate the expression:

$$\frac{P(A) - P(A \cap B)}{1 - P(B)}$$

A. $P(A \mid \bar{B})$
B. $P(A \mid B)$
C. $P(\bar{A} \mid B)$
D. $P(\bar{A} \mid \bar{B})$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #21 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Calculus

Let

$$f(x) = x^3 - 6x^2 + 9x + 7$$

defined on the interval $[-2, 6]$.
What are the global maximum and minimum values of $f(x)$ on this interval?

A. Global max $= 61$ at $x = 6$, Global min $= -43$ at $x = -2$
B. Global max $= 11$ at $x = 1$, Global min $= -43$ at $x = -2$
C. Global max $= 7$ at $x = 3$, Global min $= -43$ at $x = -2$
D. Global max $= 61$ at $x = 6$, Global min $= 7$ at $x = 3$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

**Q #22** Numerical Type | Award: 2 | Penalty: 0 | Calculus

Let

$$f(x) = x^4 + x + 3 \quad \text{for} \quad 0 \le x \le 2.$$

Find a point $c$ whose existence is guaranteed by the Mean Value Theorem. The point $c$ can be expressed as:

$$c = 2^p \quad \text{where} \quad p =$$

Your Answer: | Correct Answer: 0.33 | Not Attempted | Time taken: 00min 00sec | Discuss

**Q #23** Multiple Select Type | Award: 2 | Penalty: 0 | Linear Algebra

Let $u = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \in \mathbb{R}^3$, and define the matrix $A = uu^T$.

Which of the following statements is/are correct?

A. $\begin{bmatrix} 1 & 2 & 3 \end{bmatrix}^T$ is an eigenvector of $A$
B. $A$ has only one nonzero eigenvalue, which is $14$
C. $\text{Rank}(A) = 1$
D. Matrix $A$ has three equal eigenvalues, all equal to $14$

Your Answer: | Correct Answer: A;B;C | Not Attempted | Time taken: 00min 00sec | Discuss

**Q #24** Multiple Choice Type | Award: 2 | Penalty: 0.67 | Linear Algebra

Let $A = LU$, where $L$ is lower triangular with 1's on the main diagonal, and $U = (u_{i,j})$ is upper triangular, and

$$A = \begin{pmatrix} 2 & 1 & 3 & 0 \\ -4 & 0 & -5 & 0 \\ 2 & 5 & 5 & 2 \end{pmatrix}$$

What's $u_{2,4}$ ?

A. $2$
B. $-1$
C. $1$
D. $0$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

**Q #25** Multiple Select Type | Award: 2 | Penalty: 0 | Linear Algebra

For which of the following values of $a$ are the vectors

$$\mathbf{u} = \begin{bmatrix} 1 \\ a \\ 4 \end{bmatrix} \quad \text{and} \quad \mathbf{v} = \begin{bmatrix} 3a \\ -1 \\ -6 \end{bmatrix}$$

orthogonal?

A. $6$
B. $8$
C. $12$
D. $24$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #26 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

A fair $3$ -sided die (with faces labeled $1, 2, 3$ ) is rolled three times. Let the random variable $X$ be the sum of the first two rolls, and let $Y$ be the sum of all three rolls.
What is the value of $\mathbb{P}(X = 5 \mid Y = 7)$ ?

A. $\frac{2}{9}$

B. $\frac{1}{3}$

C. $\frac{2}{27}$

D. $\frac{2}{6}$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #27 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

Let $X$ be a discrete random variable that takes exactly two distinct values, has mean $10$, and variance $9$ .
Assume both values are taken with equal probability.
Which pair of values does the random variable $X$ take?

A. $\{7, 13\}$

B. $\{8, 12\}$

C. $\{6, 14\}$

D. $\{5, 15\}$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #28 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

An unbiased coin is tossed six times in a row and four different such trials are conducted. One trial implies six tosses of the coin. If $H$ stands for head ans $T$ stands for tail, the following are the observations from the four trials.

A. $HTHTHT$

B. $TTHHHT$

C. $HTTHHT$

D. $HHHT__$

Which statement describing the last two coin tosses of the fourth trial has the highest probability of being correct?

A. Two $T$ will occur.

B. One $H$ and one $T$ will occur.

C. Two $H$ will occur.

D. One $H$ will be followed by one $T$ .

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 02sec | Discuss

Q #29 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

Let a random variable $X$ follow a Poisson distribution such that

$$\text{Prob}(X = 1) = \text{Prob}(X = 2)$$

The value of $\text{Prob}(X = 3)$ is:

A. $\frac{4e^{-2}}{3}$

B. $\frac{9e^{-3}}{2}$

C. $\frac{2e^{-2}}{3}$

D. $\frac{3e^{-4}}{4}$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #30 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

Let $A$, $B$, and $C$ be three independent events.
Define:
$\alpha = \Pr(A)$
$\beta = \Pr($ not all three events occur simultaneously $)$
$\gamma = \Pr($ at least one of the three events does not occur $)$
$\delta = \Pr(C$ occurs, but neither A nor B occur)
Then, the probability that $C$ occurs equals:

A. $\delta/(\beta + \delta)$

B. $\beta/(\beta + \delta)$

C. $\alpha\gamma(\beta + \delta)$

D. $\gamma/(\beta + \delta)$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

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